MILITARY SPECIFICATION SHEET

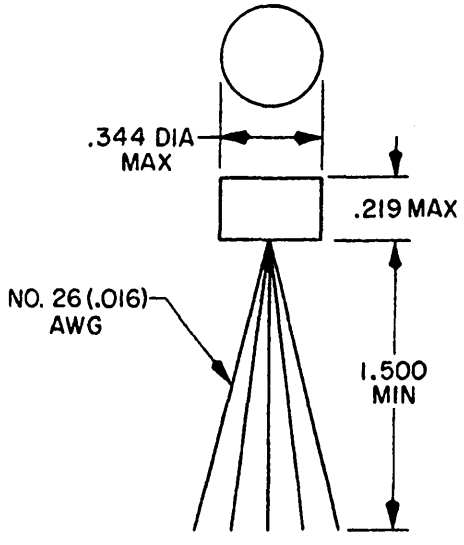
TRANSFORMERS, PULSE, LOW POWER,

TYPE TP6RX4410CZ

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-21038.

(B) This specification is inactive for new design after the date of this specification.




| INCHES | MM |
|---|---|
| .016 | .41 |
| .219 | 5.56 |
| .344 | 8.74 |
| 1.500 | 38.10 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.032 (.81 mm).

FIGURE 1. Dimensions and configuration.

(B) denotes changes

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TABLE I. Electrical characteristics including waveform parameters.

| Dash no. | DCR (ohms) | | | Blocking oscillator pulse | | | | | Coupling circuit characteristics | | | | | | | Frequency response 2/ ± 2 dB | Turns ratio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Pulse width (µs) | Rise time (µs) | Over-shoot % | Droop % | Back swing % | Pulse width (µs) | Volts out | Rise time (µs) | Over-shoot % | Droop % | Back swing % | Impedance 1/ out, in (ohms) | | |
| | 1-2 | 3-4 | 5-6 | | | | | | | | | | | | | | |
| 001 | 13.1 | 15.9 | 4.7 | 10 | 0.35 | 0 | 5 | 12 | 10 | 6.5 | 0.4 | 0 | 15 | 20 | 200 | 1.5 kHz to 1.14 MHz | 4:4:1 |
| 002 | 8.0 | 9.7 | 2.9 | 5 | 0.30 | 0 | 0 | 3 | 5 | 7.9 | 0.22 | 0 | 13 | 25 | 200 | 1.8 kHz to 1.45 MHz | 4:4:1 |
| 003 | 4.9 | 6.05 | 1.8 | 3 | 0.20 | 0 | 0 | 14 | 3 | 6.8 | 0.20 | 0 | 2 | 10 | 100 | 1.7 kHz to 1.5 MHz | 4:4:1 |
| 004 | 3.0 | 3.7 | 1.1 | 2 | 0.10 | 0 | 0 | 15 | 2 | 6.6 | 0.18 | 0 | 2 | 10 | 100 | 2.2 kHz to 1.32 MHz | 4:4:1 |
| 005 | 2.45 | 3.1 | 0.9 | 1 | 0.08 | 0 | 0 | 14 | 1 | 6.8 | 0.15 | 0 | 0 | 5 | 100 | 7.5 kHz to 2.25 MHz | 4:4:1 |
| 006 | 1.5 | 1.85 | 0.54 | 0.5 | 0.05 | 0 | 0 | 15 | 0.5 | 7 | 0.06 | 0 | 0 | 0 | 100 | 7 kHz to 3.25 MHz | 4:4:1 |
| 007 | 1.01 | 1.25 | 0.37 | 0.2 | 0.030 | 2 | 0 | 15 | 0.2 | 7 | 0.035 | 0 | 0 | 5 | 100 | 16 kHz to 9.5 MHz | 4:4:1 |
| 008 | 0.47 | 0.56 | 0.17 | 0.1 | 0.025 | 0 | 0 | 25 | 0.1 | 8 | 0.02 | 0 | 0 | 5 | 50 | 100 kHz to 17 MHz | 4:4:1 |
| 009 | 0.21 | 0.23 | 0.13 | 0.05 | 0.02 | 0 | 0 | 37 | 0.05 | 9 | 0.018 | 0 | 0 | 12 | 50 | 150 kHz to 29 MHz | 4:4:1 |

1/ See figure 2.
2/ Per coupling circuit (Z), in/out, 1 V input.





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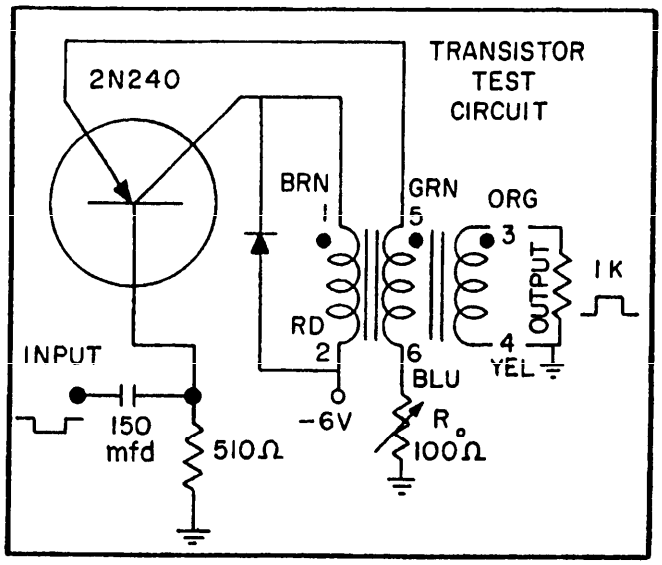



FIGURE 2. Test circuit.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal.

Terminals: Insulated, gold plated dumet leads, (see figure 2).

Weight: 0.07 ounce, maximum.

Terminal strength: Method 211, MIL-STD-202, test condition A - pull 2.5 pounds.

Dielectric withstanding voltage:

At atmospheric pressure: Method 301, MIL-STD-202. 100 volts rms for 60 seconds between windings and case.

Barometric pressure (reduced): Method 301, MIL-STD-202 and method 105, test condition E, MIL-STD-202. 100 volts rms between windings and case.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Not less than 10,000 megohms.

Electrical characteristics: See table I.

Test circuit: See figure 2.



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Life (expectancy "X"): Applicable.

Temperature rise: Not applicable.

Vibration high frequency: Method 204, test condition B, MIL-STD-202.

Shock (specified pulse): Method 213, test condition H, MIL-STD-202.

Ⓑ Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Groups A and B tests of MIL-T-21038 shall be applicable.

Part number: M21038/6 - (dash number from table I).

Preparing Activity:
Air Force - 85

(Project 5950-0587-03)



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Personnel Qualification/Training Records